Nai-Wen Chi

San Diego, California | naiwen.amanda.chi@gmail.com
Research portfolio | Google Scholar | ORCID: 0009-0006-4289-9191 | LinkedIn

RESEARCH INTERESTS

Biomaterials and drug delivery for tissue repair; material-cell interactions; targeted delivery and functional evaluation of genetic medicines.

EDUCATION

M.S., Biomedical Engineering
Feb 2015 - Jan 2017

National Tsing Hua University, Taiwan

GPA: 4.05/4.30. Thesis research: nitric-oxide-releasing microparticles for osteoporosis therapy.

B.S., Agronomy
Sep 2008 - Jun 2012

National Chung Hsing University, Taiwan

GPA: 3.25/4.00.

RESEARCH & TRANSLATIONAL EXPERIENCE

R&D and Regulatory Scientist
Dec 2025 - Present

Coastar Therapeutics, California

- Support regulatory gap analysis and evaluation of development evidence for therapeutic programs.

Research Assistant
Apr 2025 - Nov 2025

Academia Sinica, Taiwan | Dr. Yu-Jung Lin

- Prepared polymer conjugates and ligand-modified nanoparticles; adjusted formulation and purification conditions for drug-delivery research.
- Measured particle size and zeta potential and quantified drug loading by HPLC; prepared alginate hydrogels and evaluated gelation time.

R&D Engineer
Jan 2018 - Mar 2021

BioGend Therapeutics, Taiwan

- Developed tyramine-alginate hydrogel systems for cartilage adhesion and regeneration; co-authored the 2022 International Journal of Biological Macromolecules paper listed on page 2.
- Contributed to cartilage-repair implant development through scaffold fabrication and mechanical testing; supported mini-pig studies and surgical usability evaluation.
- Provided technical support during human clinical surgical cases, including equipment setup and instrument sterilization coordination; contributed to regulatory documentation.
- Named co-inventor on cartilage filling, cartilage measurement, and dry-heating device patents and applications.

R&D Engineer
May 2017 - Dec 2017

Exactech Taiwan Ltd., Taiwan

- Supported R&D projects and clinical trials for Class III medical devices. The Taiwan operation was acquired by BioGend in January 2018.

Master's Research | Nitric Oxide Delivery for Osteoporosis
2015 - 2017

National Tsing Hua University, Taiwan

- Developed phase-change delivery systems for nitric oxide (NO); performed in vitro assays and in vivo studies using an ovariectomized rat model of osteoporosis.
- Contributed to the 2018 Advanced Materials paper and was named co-inventor on a temperature-responsive delivery composition patent family.

PEER-REVIEWED PUBLICATIONS

1. Wong CC, Lu CX, Cho EC, Lee PW, **Chi NW**, Lin PY, Jheng PR, Chen HL, Mansel BW, Chen YM, Chen CH, Chuang EY. Calcium peroxide aids tyramine-alginate gel to crosslink with tyrosinase for efficient cartilage repair. *International Journal of Biological Macromolecules*. 2022;208:299-313. doi:10.1016/j.ijbiomac.2022.03.044

2. Lin YJ, Chen CC, **Chi NW**, Nguyen T, Lu HY, Nguyen D, Lai PL, Sung HW. In Situ Self-Assembling Micellar Depots that Can Actively Trap and Passively Release NO with Long-Lasting Activity to Reverse Osteoporosis. *Advanced Materials*. 2018;30(22):1705605. doi:10.1002/adma.201705605

PATENTS & PUBLISHED APPLICATIONS | CO-INVENTOR

Temperature-responsive composition, method for fabricating, and use thereof. National Tsing Hua University. Taiwan invention patent TWI606829B (2017; subsequently lapsed for non-payment of fees). U.S. published application US20180161276A1 (2018; abandoned).

Cartilage filling system and surgical instrument set including the same. BioGend Therapeutics. Utility models TWM594960U (2020) and CN212730081U (2021). U.S. counterpart, published as US20210252227A1 (2021), abandoned.

Cartilage volumetric device and surgical instrument set including the same. BioGend Therapeutics. Utility models TWM594437U (2020) and CN212729944U (2021; English title: Cartilage quantifier and surgical instrument set).

Dry type heating device. BioGend Therapeutics. Utility models TWM590460U (2020) and CN211454393U (2020).

REGULATORY EXPERIENCE

Senior Regulatory Affairs Specialist
May 2022 - Jul 2023

Winnoz Technology, Taiwan

- Led TFDA medical-device license applications and amendments; prepared EU IVDR and U.S. FDA IVD submission documentation and supported ISO 13485 and TFDA QMS audits.

Regulatory Affairs Specialist / Project Manager
Apr 2021 - May 2022

BioGend Therapeutics, Taiwan

- Managed TFDA Class I-III device applications; led a Class II application and coordinated supporting preclinical evidence. Supported design controls, ISO 13485 and TFDA QMS audits, and international documentation including EU MDR requirements.

RESEARCH METHODS

Materials & delivery: Polymer conjugation and ligand modification; nanoparticle and liposome preparation; scaffold and alginate-hydrogel fabrication.

Characterization: DLS and zeta potential; HPLC drug-loading analysis; gelation-time and mechanical testing; sample preparation for NMR and GPC.

Biological evaluation: In vitro assays; ovariectomized rat osteoporosis studies; support for mini-pig studies. WST-1, ALP activity, Alizarin Red S, H&E histology, serum ALP and osteocalcin.

ADDITIONAL EXPERIENCE & PROFESSIONAL DEVELOPMENT

Chief Executive Officer & Project Manager, Wen Fu AI Co., Taiwan (Jul 2023 - Mar 2024). Founded a small technology venture; managed online-course and AI-assisted website projects, including scope, schedules, and client communication.

Graduate study preparation, Taiwan (Mar 2024 - Mar 2025). Prepared doctoral applications and completed English proficiency testing.

Languages: Mandarin (native); English (professional working proficiency).